1) gcc ex2.c

lab08 > examples > C ex2.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

int main()
{
    printf("%d\n", getpid());

    char *msg = malloc(100 * sizeof(char));
    strcpy(msg, "eeeeh what am i doing....?\n");

    while (1) sleep(1);
    free(msg);
}
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL CODE REFERENCE LOG

```
[me@yoshi examples]$ gcc ex2.c
[me@yoshi examples]$ ./a.out
561455
```

2) cat /proc/561455/maps | grep heap

lab08 > examples > C ex2.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

int main()
{
    printf("%d\n", getpid());

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    strcpy(msg, "eeeeh what am i doing....?\n");

    while (1) sleep(1);
    free(msg);
}
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL CODE REFERENCE LOG

```
~/i/o/osy ❱❱❱ bash
[me@yoshi osy]$
[me@yoshi osy]$ cat /proc/561455/maps | grep heap
55bff64c2000-55bff64e3000 rw-p 00000000 00:00 0                          [heap]
[me@yoshi osy]$
```

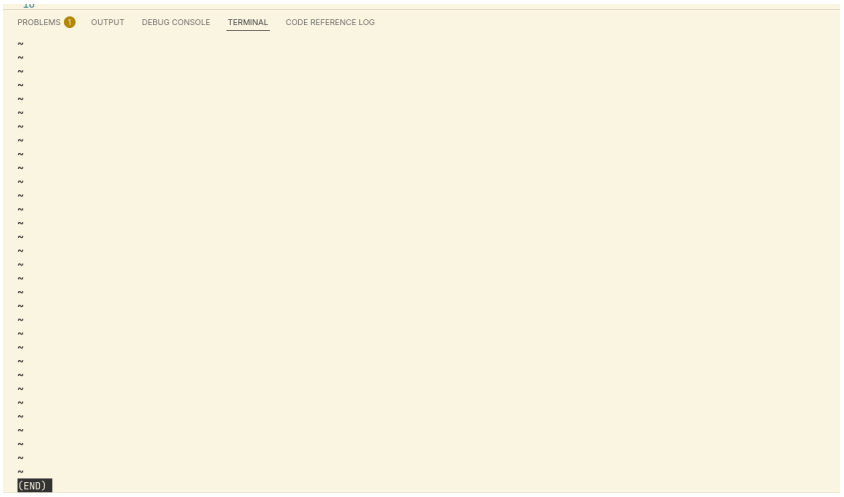
Heap:
start = 0x55bff64c2000
end = 0x55bff64e3000
size = 0x21000 = 135168 bytes = 132 KB.

3) sudo xxd -s 0x55bff64c2000 -l 135168 /proc/561455/mem | less

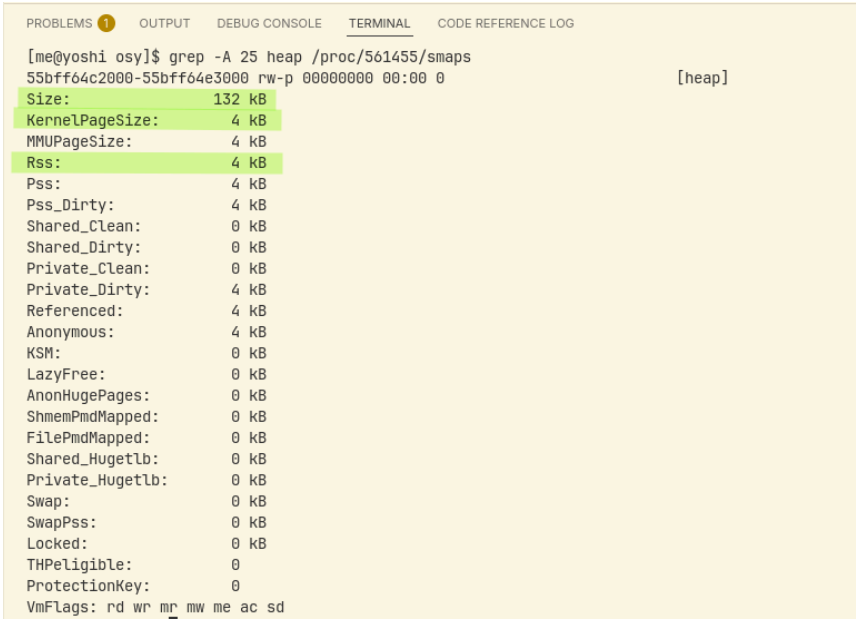
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL CODE REFERENCE LOG

```
55bff64c23e0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c23f0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c2400: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c2410: 0000 0000 0000 0000 7100 0000 0000 0000  ........q.......
55bff64c2420: 6565 6565 6820 7768 6174 2061 6d20 6920  eeeeh what am i
55bff64c2430: 646f 696e 672e 2e2e 2e3f 0a00 0000 0000  doing....?......
55bff64c2440: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c2450: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c2460: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c2470: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c2480: 0000 0000 0000 0000 810b 0200 0000 0000  ................
55bff64c2490: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c24a0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c24b0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c24c0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c24d0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c24e0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c24f0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
55bff64c2500: 0000 0000 0000 0000 0000 0000 0000 0000  ................
```

4) sudo less /proc/561455/mem



5) grep -A 25 heap /proc/561455/smaps



(132 / 4 = 33 pages total)

6) sudo ls -l /proc/561455/map_files/

```
PROBLEMS 1  OUTPUT  DEBUG CONSOLE  TERMINAL  CODE REFERENCE LOG

[me@yoshi osy]$ sudo ls -l /proc/561455/map_files/
total 0
lr-------- 1 me me 64 Oct 28 18:06 55bfe247b000-55bfe247c000 -> /home/me/inno/osy/osy/lab08/examples/a.out
lr-------- 1 me me 64 Oct 28 18:06 55bfe247c000-55bfe247d000 -> /home/me/inno/osy/osy/lab08/examples/a.out
lr-------- 1 me me 64 Oct 28 18:06 55bfe247d000-55bfe247e000 -> /home/me/inno/osy/osy/lab08/examples/a.out
lr-------- 1 me me 64 Oct 28 18:06 55bfe247e000-55bfe247f000 -> /home/me/inno/osy/osy/lab08/examples/a.out
lr-------- 1 me me 64 Oct 28 18:06 55bfe247f000-55bfe2480000 -> /home/me/inno/osy/osy/lab08/examples/a.out
lr-------- 1 me me 64 Oct 28 18:06 7f3a9f400000-7f3a9f424000 -> /usr/lib/libc.so.6
lr-------- 1 me me 64 Oct 28 18:06 7f3a9f424000-7f3a9f59f000 -> /usr/lib/libc.so.6
lr-------- 1 me me 64 Oct 28 18:06 7f3a9f59f000-7f3a9f615000 -> /usr/lib/libc.so.6
lr-------- 1 me me 64 Oct 28 18:06 7f3a9f615000-7f3a9f619000 -> /usr/lib/libc.so.6
lr-------- 1 me me 64 Oct 28 18:06 7f3a9f619000-7f3a9f61b000 -> /usr/lib/libc.so.6
lr-------- 1 me me 64 Oct 28 18:06 7f3a9f799000-7f3a9f7b4000 -> '/etc/ld.so.cache (deleted)'
lr-------- 1 me me 64 Oct 28 18:06 7f3a9f7bc000-7f3a9f7bd000 -> /usr/lib/ld-linux-x86-64.so.2
lr-------- 1 me me 64 Oct 28 18:06 7f3a9f7bd000-7f3a9f7ea000 -> /usr/lib/ld-linux-x86-64.so.2
lr-------- 1 me me 64 Oct 28 18:06 7f3a9f7ea000-7f3a9f7f9000 -> /usr/lib/ld-linux-x86-64.so.2
lr-------- 1 me me 64 Oct 28 18:06 7f3a9f7f9000-7f3a9f7fb000 -> /usr/lib/ld-linux-x86-64.so.2
lr-------- 1 me me 64 Oct 28 18:06 7f3a9f7fb000-7f3a9f7fc000 -> /usr/lib/ld-linux-x86-64.so.2
[me@yoshi osy]$
```

7) pmap 561455

```
[me@yoshi osy]$ pmap 561455
561455:   ./a.out
000055bfe247b000      4K r---- a.out
000055bfe247c000      4K r-x-- a.out
000055bfe247d000      4K r---- a.out
000055bfe247e000      4K r---- a.out
000055bfe247f000      4K rw--- a.out
000055bff64c2000    132K rw---   [ anon ]
00007f3a9f400000    144K r---- libc.so.6
00007f3a9f424000   1516K r-x-- libc.so.6
00007f3a9f59f000    472K r---- libc.so.6
00007f3a9f615000     16K r---- libc.so.6
00007f3a9f619000      8K rw--- libc.so.6
00007f3a9f61b000     32K rw---   [ anon ]
00007f3a9f794000     20K rw---   [ anon ]
00007f3a9f799000    108K r---- ld.so.cache (deleted)
00007f3a9f7b4000     16K r----   [ anon ]
00007f3a9f7b8000      8K r----   [ anon ]
00007f3a9f7ba000      8K r-x--   [ anon ]
00007f3a9f7bc000      4K r---- ld-linux-x86-64.so.2
00007f3a9f7bd000    180K r-x-- ld-linux-x86-64.so.2
00007f3a9f7ea000     60K r---- ld-linux-x86-64.so.2
00007f3a9f7f9000      8K r---- ld-linux-x86-64.so.2
00007f3a9f7fb000      4K rw--- ld-linux-x86-64.so.2
00007f3a9f7fc000      4K rw---   [ anon ]
00007ffff3dd8000    132K rw---   [ stack ]
ffffffffff600000      4K --x--   [ anon ]
 total             2896K
```

```
[me@yoshi osy]$ pmap -x 561455
561455:   ./a.out
Address           Kbytes     RSS   Dirty Mode  Mapping
000055bfe247b000       4       4       0 r---- a.out
000055bfe247c000       4       4       0 r-x-- a.out
000055bfe247d000       4       4       0 r---- a.out
000055bfe247e000       4       4       4 r---- a.out
000055bfe247f000       4       4       4 rw--- a.out
000055bff64c2000     132       4       4 rw---   [ anon ]
00007f3a9f400000     144     144       0 r---- libc.so.6
00007f3a9f424000    1516    1192       0 r-x-- libc.so.6
00007f3a9f59f000     472     144       0 r---- libc.so.6
00007f3a9f615000      16      16      16 r---- libc.so.6
00007f3a9f619000       8       8       8 rw--- libc.so.6
00007f3a9f61b000      32      20      20 rw---   [ anon ]
00007f3a9f794000      20      12      12 rw---   [ anon ]
00007f3a9f799000     108     108       0 r---- ld.so.cache (deleted)
00007f3a9f7b4000      16       0       0 r----   [ anon ]
00007f3a9f7b8000       8       0       0 r----   [ anon ]
00007f3a9f7ba000       8       8       0 r-x--   [ anon ]
00007f3a9f7bc000       4       4       0 r---- ld-linux-x86-64.so.2
00007f3a9f7bd000     180     180       0 r-x-- ld-linux-x86-64.so.2
00007f3a9f7ea000      60      60       0 r---- ld-linux-x86-64.so.2
00007f3a9f7f9000       8       8       8 r---- ld-linux-x86-64.so.2
00007f3a9f7fb000       4       4       4 rw--- ld-linux-x86-64.so.2
00007f3a9f7fc000       4       4       4 rw---   [ anon ]
00007ffff3dd8000     132      12      12 rw---   [ stack ]
ffffffffff600000       4       0       0 --x--   [ anon ]
---------------- ------- ------- -------
total kB            2896    1948      96
```

8) free -t -h

```
PROBLEMS 1  OUTPUT  DEBUG CONSOLE  TERMINAL  CODE REFERENCE LOG

[me@yoshi osy]$ free -t -h
               total        used        free      shared  buff/cache   available
Mem:           7.4Gi       4.1Gi       945Mi       831Mi       3.5Gi       3.3Gi
Swap:             0B          0B          0B
Total:         7.4Gi       4.1Gi       945Mi
[me@yoshi osy]$
```

9) vmstat 1 5

```
[me@yoshi osy]$ vmstat 1 5
procs -----------memory---------- ---swap-- -----io---- -system-- -------cpu-------
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st gu
 1  0      0 1078476 235304 3443488  0    0    28   503  883    4  2  1 96  0  0  0
 0  0      0 1085448 235312 3431416  0    0     0    72 4064 5419  6  6 89  0  0  0
 1  0      0 1084764 235312 3427348  0    0     0     0  905 2033  1  1 98  0  0  0
 0  0      0 1084860 235312 3427252  0    0     0     0  749 1675  2  2 96  0  0  0
 0  0      0 1104848 235312 3425752  0    0     0     0 1579 1666 16  1 83  0  0  0
[me@yoshi osy]$
```

10) top -d 1

```
top - 18:16:14 up 1 day,  6:00,  1 user,  load average: 0.89, 0.74, 0.61
Tasks: 234 total, 1 running, 231 sleep, 0 d-sleep, 2 stopped, 0 zombie
%Cpu(s):  5.5 us,  3.8 sy,  0.0 ni, 88.2 id,  0.0 wa,  2.2 hi,  0.2 si,  0.0 st
MiB Mem :   7619.0 total,    872.2 free,   4274.3 used,   3651.8 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   3344.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 488480 me        20   0 1409.9g 381188 123752 S   8.0   4.9   6:13.99 code
 488377 me        20   0 1393.9g 225616 141924 S   6.0   2.9   1:02.07 code
   1232 me        20   0   53.1g 486524 315044 S   5.0   6.2  10:02.75 chrome
 581420 me        20   0 1450.3g 133496 103252 S   4.0   1.7   0:00.04 chrome
 564564 root      20   0 1271652  42848  32124 S   3.0   0.5   0:27.69 sing-box
    312 root     -51   0       0      0      0 S   2.0   0.0   1:53.51 irq/137-MSFT0004:00
   1271 me        20   0   53.2g 237972 142208 S   2.0   3.1  16:53.02 chrome
 488443 me        20   0   33.0g 233532 186088 S   2.0   3.0   2:17.40 code
 516188 me        20   0 1457.8g 496464 184132 S   2.0   6.4   2:06.63 chrome
    433 root      20   0 1827032  43192  28048 S   1.0   0.6   1:57.45 containerd
   1002 me        20   0  859264 109584  60512 S   1.0   1.4   9:12.52 Xorg
 488556 me        20   0 1393.7g 126484  81440 S   1.0   1.6   1:03.76 code
 581306 me        20   0    7864   5772   3552 R   1.0   0.1   0:00.11 top
 581430 me        20   0 1448.4g  84468  60424 S   1.0   1.1   0:00.01 chrome
      1 root      20   0   20772  14352  10424 S   0.0   0.2   9:42.66 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.07 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
```

11) top -d 1 -p 561455

(our program is running in another terminal window, so it runs in parallel with top, just like a background process)

```
top - 18:17:40 up 1 day,  6:02,  1 user,  load average: 0.53, 0.67, 0.60
Tasks: 1 total, 0 running, 1 sleep, 0 d-sleep, 0 stopped, 0 zombie
%Cpu(s):  1.3 us,  1.0 sy,  0.0 ni, 96.7 id,  0.0 wa,  0.8 hi,  0.3 si,  0.0 st
MiB Mem :   7619.0 total,   1005.6 free,   4231.9 used,   3568.5 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   3387.1 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 561455 me        20   0    2892   1948   1852 S   0.0   0.0   0:00.05 a.out
```

12) getrusage()

lab08 > examples > C ex2.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

// montior memory usage of the program
#include <sys/resource.h>

int main()
{
    printf("%d\n", getpid());

    char *msg = malloc(100 * sizeof(char));
    strcpy(msg, "eeeeh what am i doing....?\n");

    struct rusage usage;
    getrusage(RUSAGE_SELF, &usage);
    printf("max RSS: %ld KB\n", usage.ru_maxrss);

    while (1) sleep(1);
    free(msg);
}
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL ···

```
[me@yoshi examples]$ gcc ex2.c
[me@yoshi examples]$ ./a.out
586562
max RSS: 1788 KB
```

./a.o
bash
bash